

Prediksi **RUL** Mesin Turbofan **ANN** Pada Dataset **C-MAPSS**

- **RUL** Remaining Useful Life
- **ANN** Artificial Neural Network
- **C-MAPSS** Commercial Modular Aero-Propulsion System Simulation

Mata Kuliah: **Data Analitik**

NPM/ Nama:

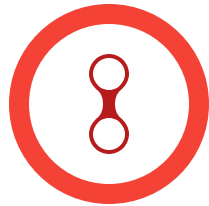
- 2506565466/ Luqman Alio Arhab
- 2506565414/ Azfa Rhazasta Abiandra
- 2506565401/ Adamsyah Haryo Saksono
- 2506565433/ Epsiarto Rizqi Nurwijayadi

Depok, Indonesia

Mei 2026

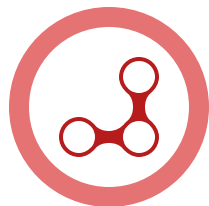


Daftar Isi



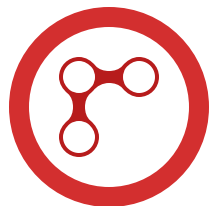
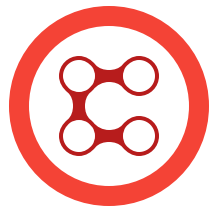
Latar Belakang

- 3 ... Pertanyaan
- 4 ... Dataset C-MAPSS
- 5 ... Struktur Mesin dan Sinyal Degradasi

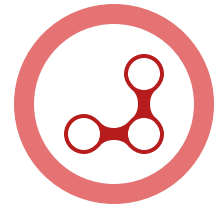


Pembersihan Data

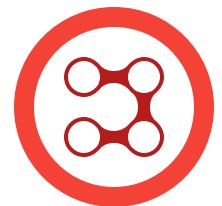
- 6 ... Persiapan Data
- 7 ... Normalisasi Z-Score
- 8 ... Seleksi Fitur: Pearson dan PCA
- 9 ... PCA: Separasi Sehat vs Terdegradasi
- 10 ... RUL Cap dan NASA Score



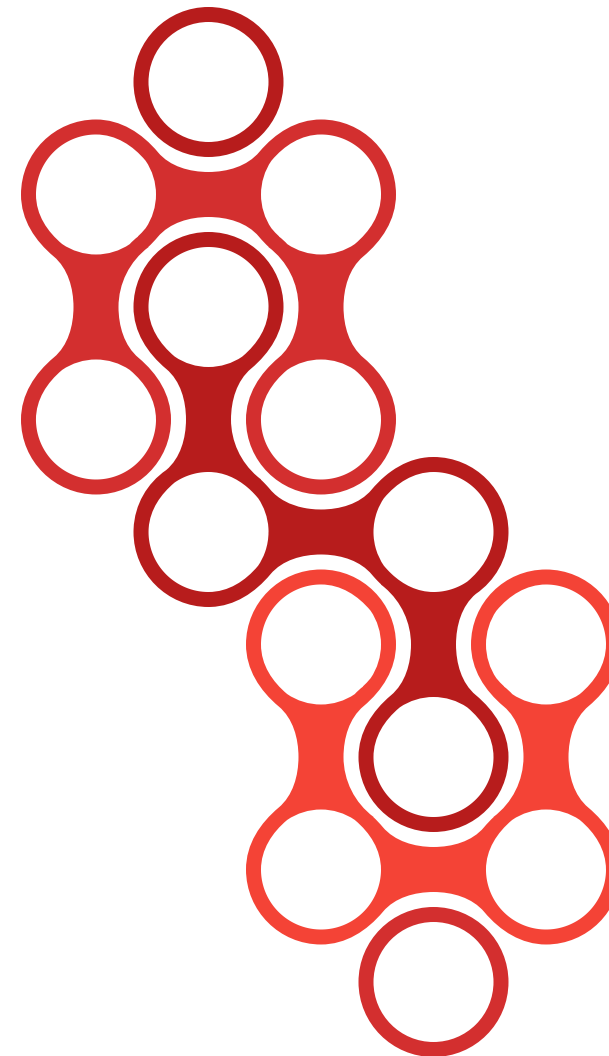
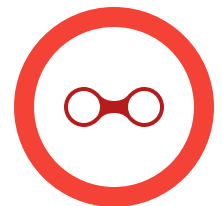
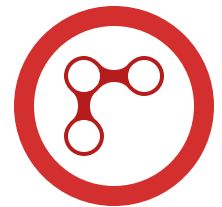
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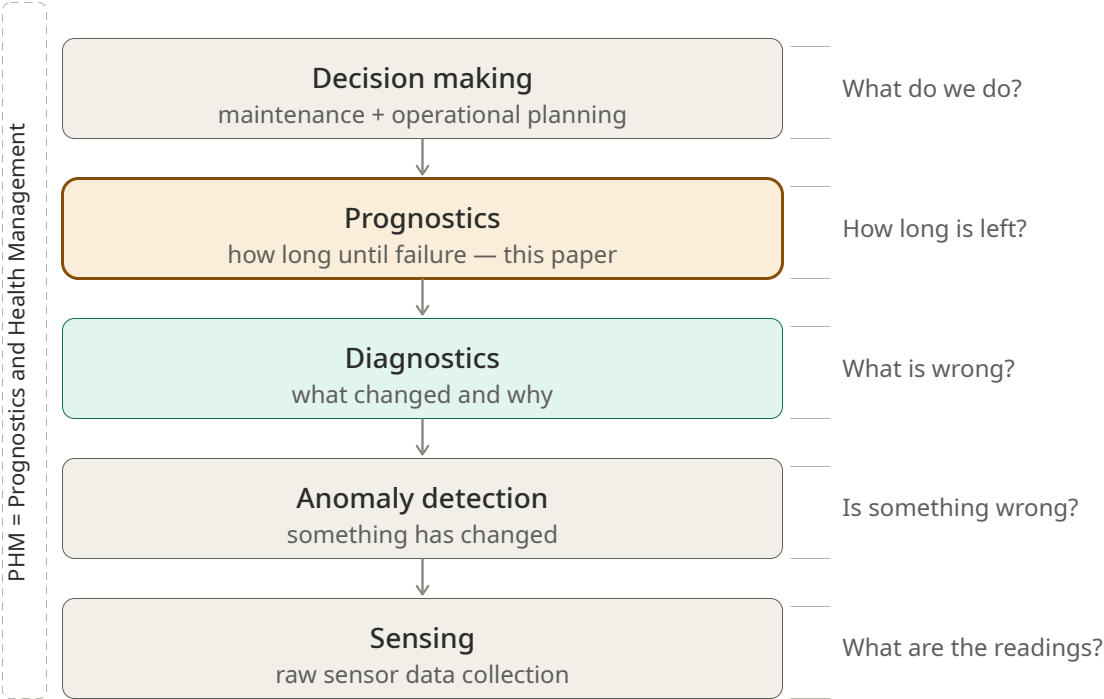
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Pertanyaan

Dari pembacaan sensor saat ini, berapa siklus tersisa sebelum mesin gagal?

Ekosistem PHM
Dari pembacaan sensor saat ini, berapa siklus tersisa sebelum mesin gagal?
Gambaran Singkat ▶ 1. RUL RUL (Remaining Useful Life) adalah sisa siklus sebelum kegagalan komponen ▶ 2. PHM Digunakan dalam sistem PHM (Prognostics and Health Management) ▶ 3. Tujuan Meng-optimalkan jadwal perawatan, kurangi downtime tak terencana ▶ 4. C-MAPSS Dataset: C-MAPSS NASA simulasi turbofan dari kondisi sehat hingga gagal.



Dataset C-MAPSS

Dataset C-MAPSS

Berfokus pada FD001, supaya paham rincian.

Sumber Data Turbofan Asli dari NASA

▶ 1. 4 sub-dataset

FD001 sampai FD004

▶ 2. Kondisi operasi

1 sampai 6

▶ 3. Mode kerusakan

HPC, atau HPC+Fan

▶ 4. 26 kolom

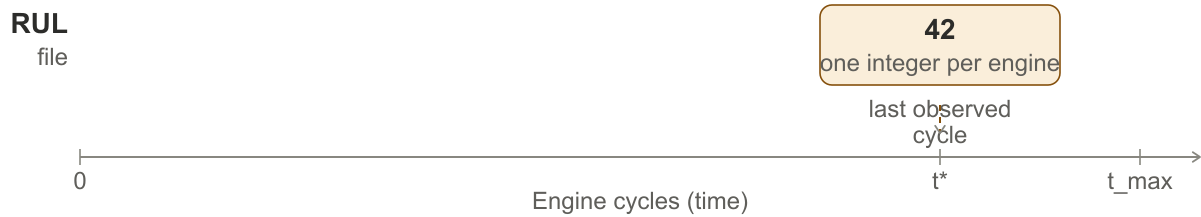
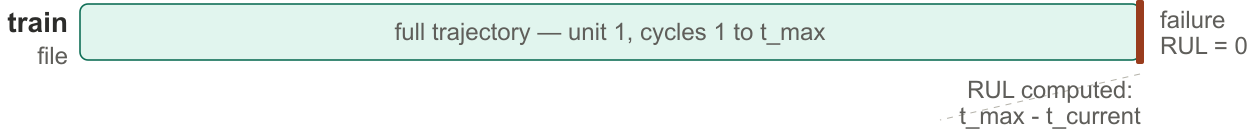
unit, cycle, 3 op_settings, 21 sensor

▶ 5. Train

Siklus penuh hingga gagal → RUL dapat dihitung

▶ 6. Test

Dipotong sebelum gagal → RUL harus diprediksi



RUL_FDxxx.txt contains one value per test engine: the true cycles remaining after the last row of that engine in the test file.

Dataset	Conditions op settings	Fault modes degraded modules	Train trajectories	Test trajectories	RUL file ground truth
FD001 easiest	1 sea level only	1 HPC degradation	100	100	RUL_FD001.txt 100 values
FD002 medium	6 alt / Mach / TRA	1 HPC degradation	260	259	RUL_FD002.txt 259 values
FD003 medium	1 sea level only	2 HPC + Fan	100	100	RUL_FD003.txt 100 values
FD004 hardest	6 alt / Mach / TRA	2 HPC + Fan	248	249	RUL_FD004.txt 249 values

increasing difficulty

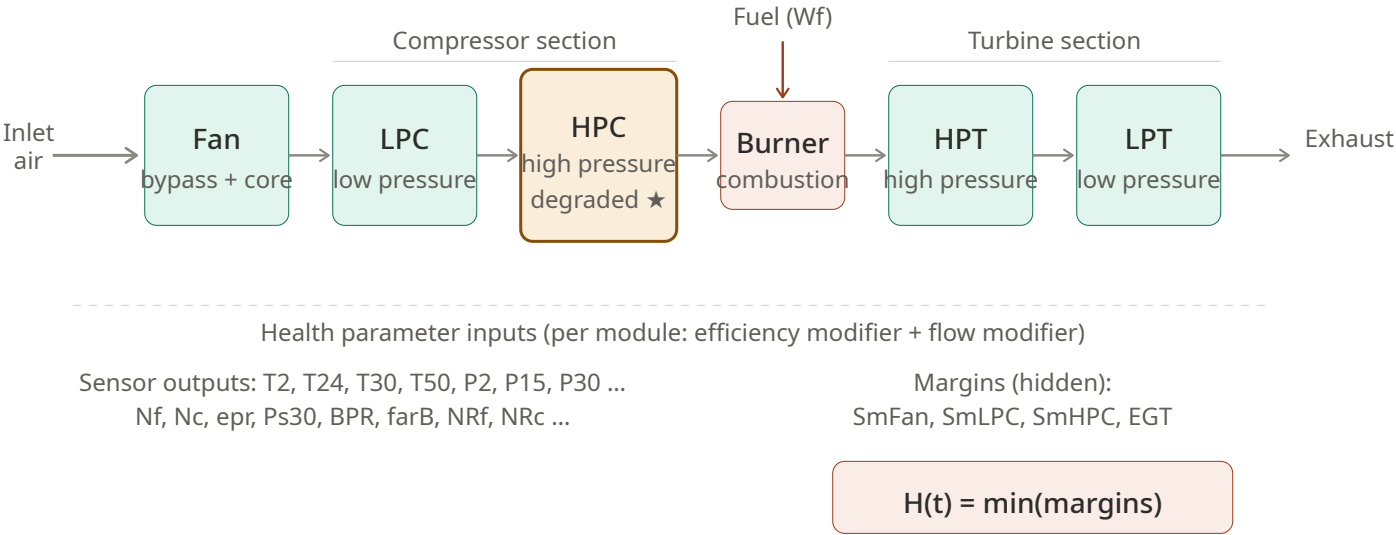
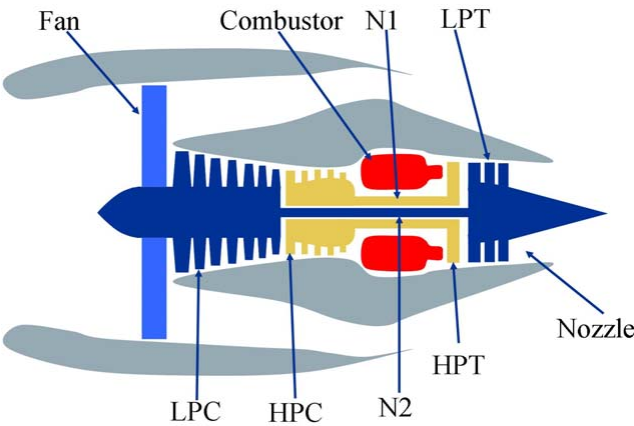
Conditions axis: FD001/FD003 (1 condition) vs FD002/FD004 (6 conditions)

Fault mode axis: FD001/FD002 (1 fault mode) vs FD003/FD004 (2 fault modes)

Files per sub-dataset: train_FDxxx.txt test_FDxxx.txt RUL_FDxxx.txt (12 files total)

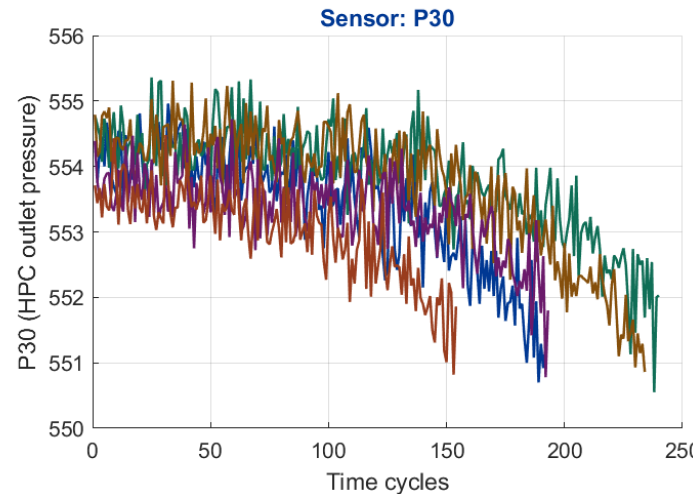
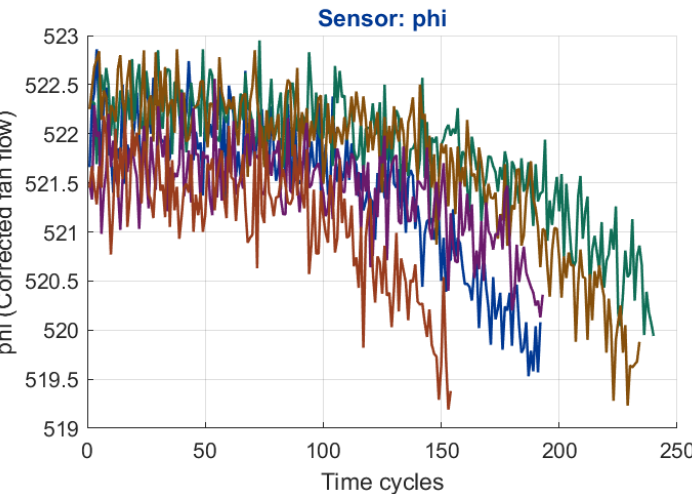
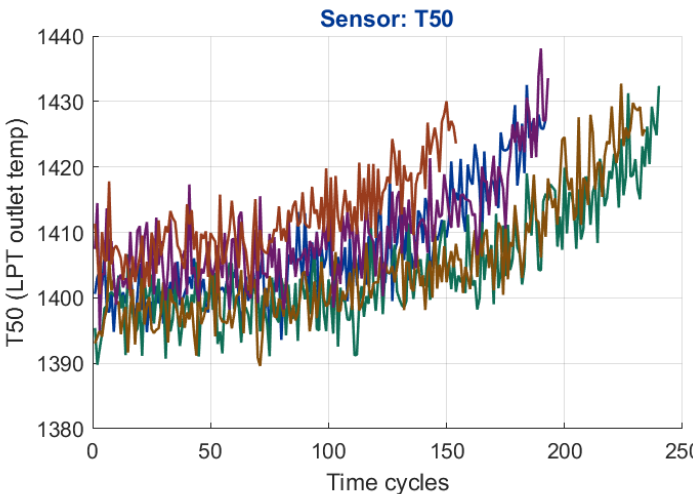
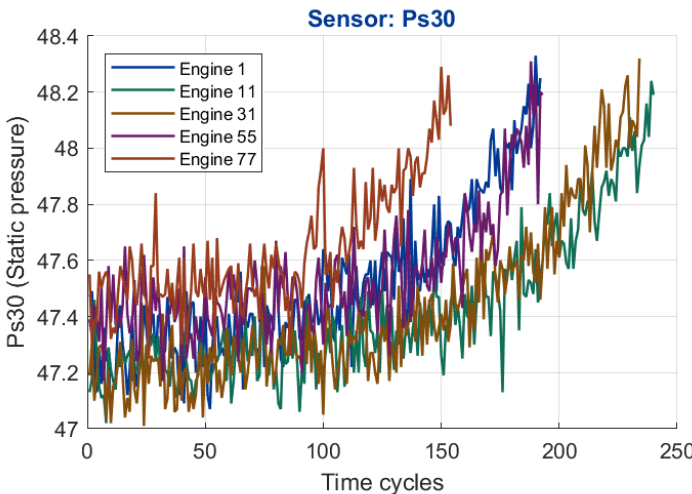
26 columns in train/test (unit, cycle, 3 op settings, 21 sensors). RUL file: one integer per engine.

Struktur Mesin dan Sinyal Degradasi



21 sensor mengukur kondisi sepanjang aliran udara

Sensor Degradation Trajectories — C-MAPSS FD001
Each line = one engine from first cycle to failure



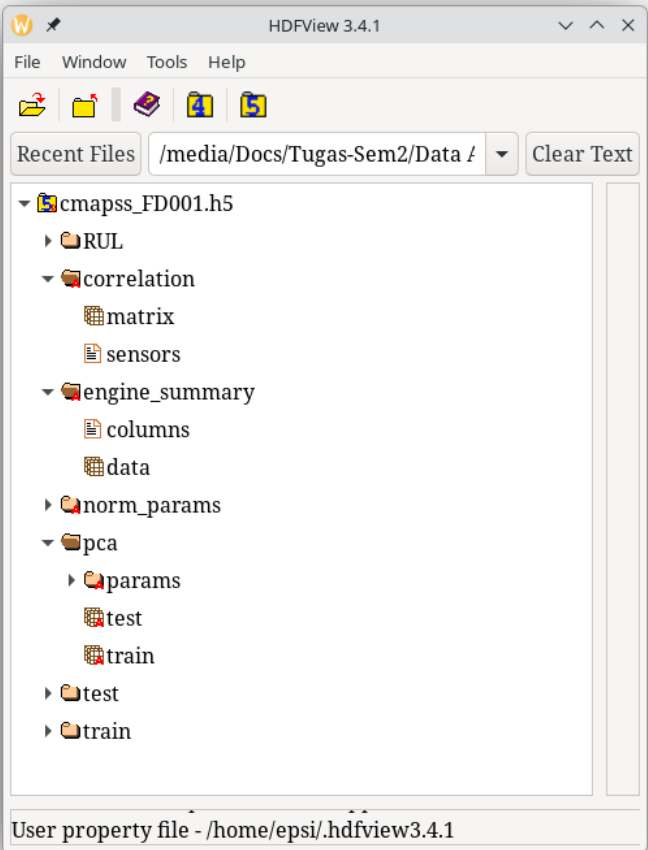
Sensor drift menjelang kegagalan

Persiapan Data: xlsx, hdf5, mat

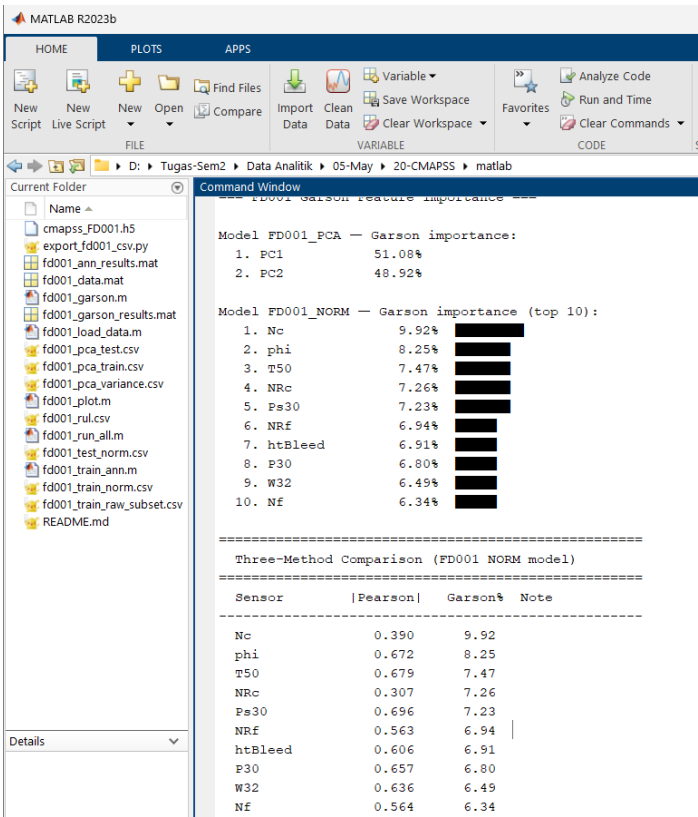
Mengingat penggunaan tools yang bebrbeda maka format artefak dokumen juga berbeda.
Data dimulai dari spreadsheet **xlsx**, lalu **hdf5** untuk python dan **mat** untuk matlab

	A	B	C	D	E	F	G	H	I	J
1	Sensor	Condition	Raw values				Normalized			
2			min	max	mean	std	min	max	mean	std
3	T2	1	518.6700	518.6700	518.6700	0.0000	0.0000	0.0000	0.0000	0.0000
4	T24	1	641.2100	644.5300	642.6809	0.5001	-3.1797	5.6963	0.7528	1.3369
5	T30	1	1571.0400	1616.9100	1590.5231	6.1311	-3.4672	6.3811	0.7159	1.3164
6	T50	1	1382.2500	1441.4900	1408.9338	9.0006	-3.5472	6.4955	0.9764	1.5258
7	P2	1	14.6200	14.6200	14.6200	0.0000	-0.9999	-0.9999	-0.9999	0.0000
8	P15	1	21.6000	21.6100	21.6098	0.0014	-4.9323	0.2027	0.1017	0.7132
9	P30	1	549.8500	556.0600	553.3677	0.8851	-6.6086	3.4823	-0.8925	1.4382
10	Nf	1	2387.9000	2388.5600	2388.0967	0.0710	-2.9642	9.3070	0.6921	1.3198
11	Nc	1	9021.7300	9244.5900	9065.2429	22.0829	-4.2214	22.2055	0.9384	2.6186
12	epr	1	1.3000	1.3000	1.3000	0.0000	0.0000	0.0000	0.0000	0.0000
13	Ps30	1	46.8500	48.5300	47.5412	0.2671	-2.9856	6.7325	1.0125	1.5450
14	phi	1	518.6900	523.3800	521.4135	0.7376	-6.4174	3.0213	-0.9364	1.4843
15	NRf	1	2387.8800	2388.5600	2388.0962	0.0719	-3.2845	9.2328	0.6944	1.3239
16	NRc	1	8099.9400	8293.7200	8143.7527	19.0762	-4.8381	19.5312	0.6717	2.3990
17	BPR	1	8.3249	8.5848	8.4421	0.0375	-3.6037	6.2853	0.8574	1.4270
18	farB	1	0.0300	0.0300	0.0300	0.0000	-0.9999	-0.9999	-0.9999	0.0000
19	htBleed	1	388.0000	400.0000	393.2107	1.5488	-3.8016	6.7401	0.7758	1.3606
20	Nf_dmd	1	2388.0000	2388.0000	2388.0000	0.0000	0.0000	0.0000	0.0000	0.0000
21	PCNfR_dmd	1	100.0000	100.0000	100.0000	0.0000	0.0000	0.0000	0.0000	0.0000
22	W31	1	38.1400	39.4300	38.8163	0.1807	-6.0468	3.9071	-0.8286	1.3947
23	W32	1	22.8942	23.6184	23.2897	0.1083	-5.8800	3.3745	-0.8259	1.3833
24										
25										

Spreadsheet: ***.xlsx**



***.hd5** untuk python



***.mat** untuk matlab

Data asli dari NASA berupa berkas text (***.txt**)

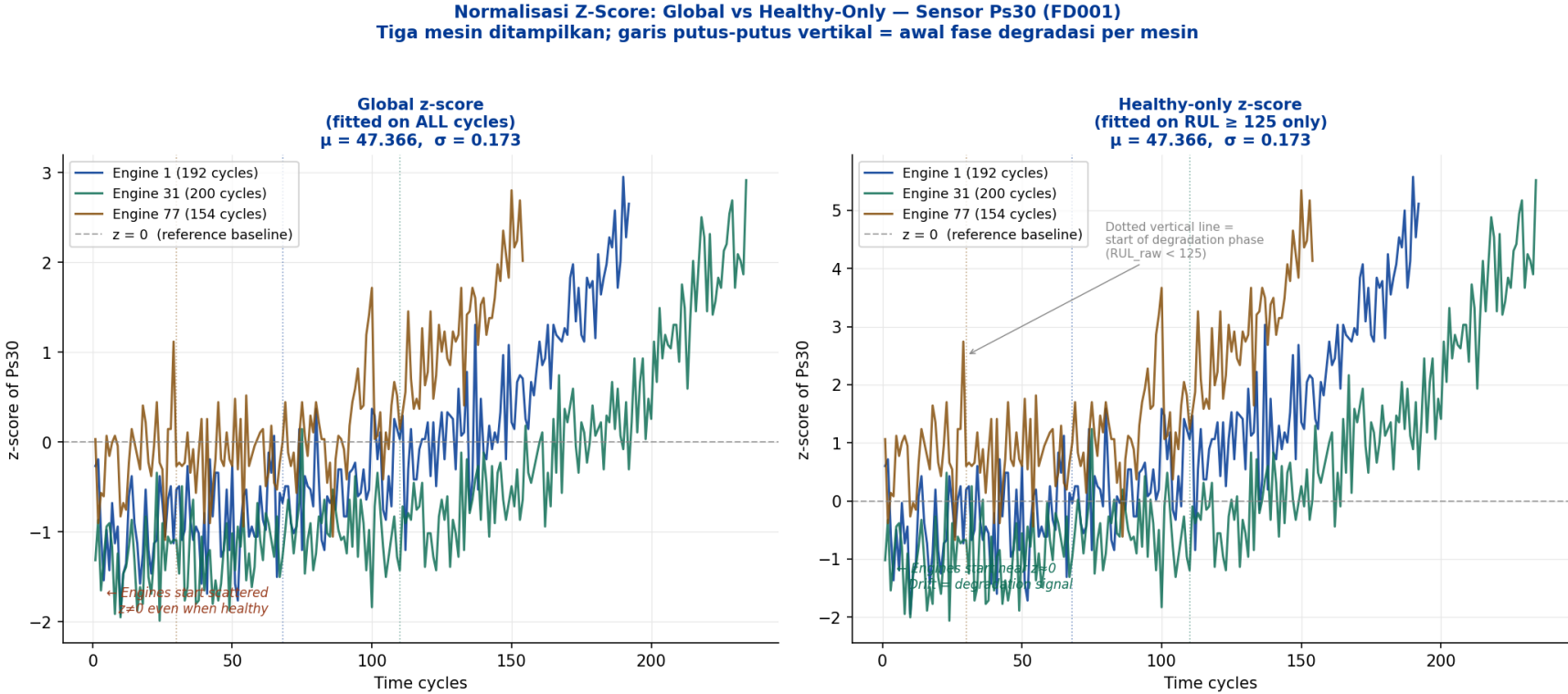
Pilihan Data Cleaning

Normalisasi Z-Score Berbasis Siklus Sehat

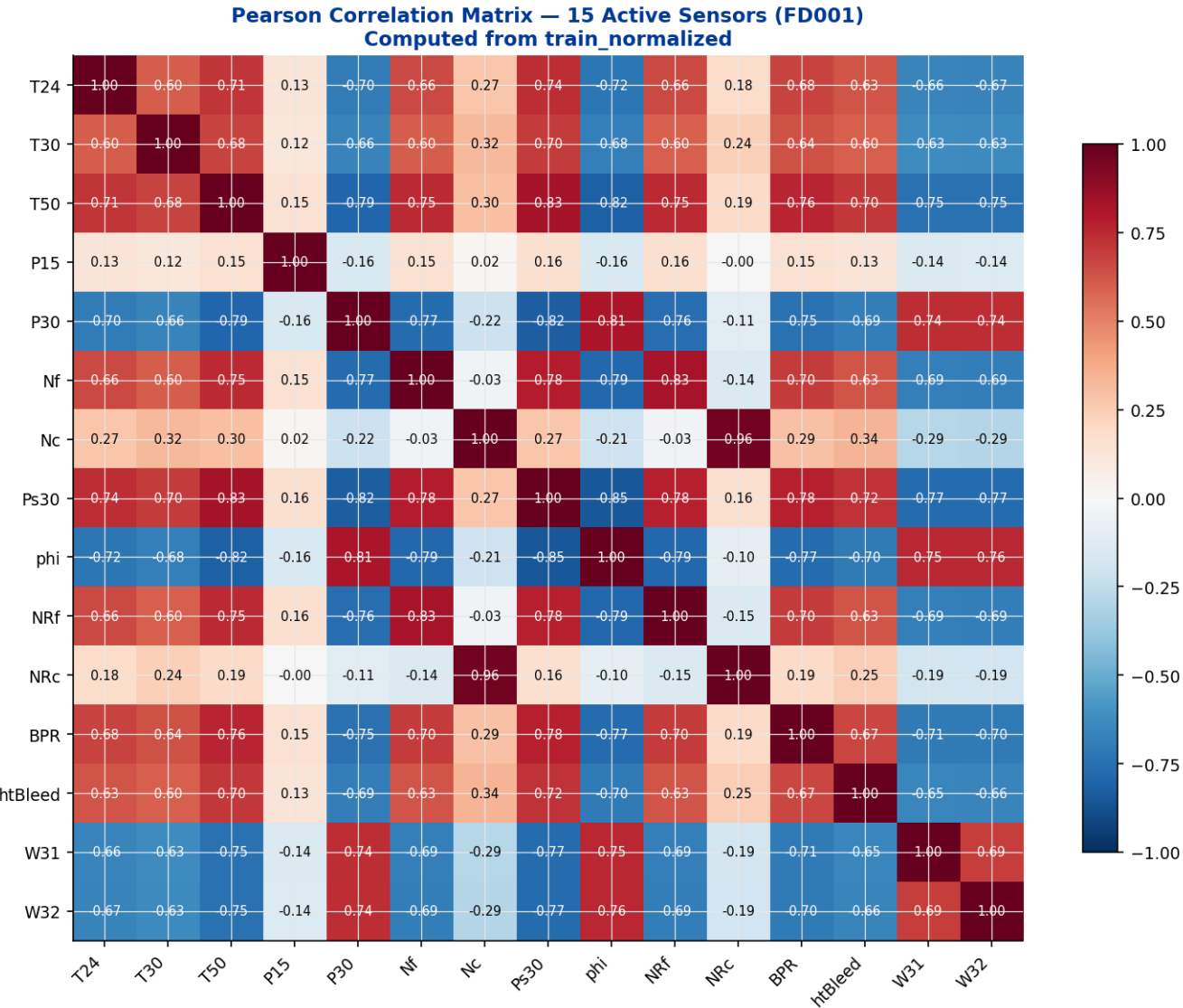
5 Metode Analisis
Pilihan Data Cleaning
▶ 1. Standar deviasi (std) ▶ 2. Koesien variasi (CV) ▶ 3. Korelasi Pearson vs RUL ▶ 4. Korelasi Spearman vs RUL ▶ 5. Mutual Information vs RUL
Sensor dieliminasi FD001
▶ 1. Alasan std = 0 → Pearson/Spearman = NaN ▶ 2. Drop T2, P2, epr, farB, Nf_{dmd}, PCNfR_{dmd}

Normalisasi
Z-Score Berbasis Siklus Sehat
▶ 1. Formula $z = \frac{x - \mu_{c, \text{healthy}}}{\sigma_{c, \text{healthy}}}$
Tiga prinsip
▶ 1. Capped Paramater Parameter dari siklus healthy (RUL ≥ 125) saja. ▶ 2. Pengelompokan Per kondisi operasi (bukan global) ▶ 3. Patokan Parameter train diterapkan ke test untuk menghindari data leakage.

Hasil
▶ z ≈ 0 = sehat ▶ z menyimpang = terdegradasi



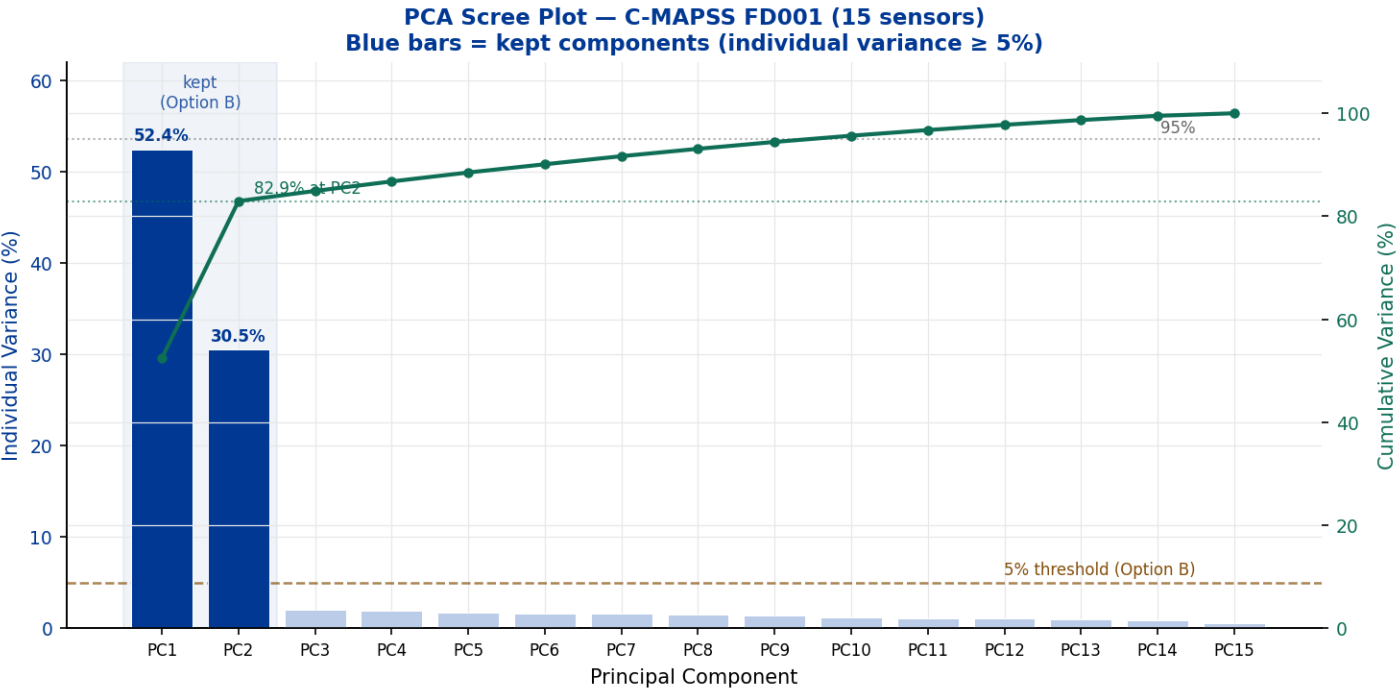
Kanan: ketiga mesin mulai di $z \approx 0$, drift seiring degradasi



Korelasi Pearson Nc-NRC $r = 0,96$
(redundansi tinggi $\Rightarrow$ motivasi PCA)

Nf = fan speed. NRf = corrected fan speed.
Pada dasarnya mengukur hal yang sama.

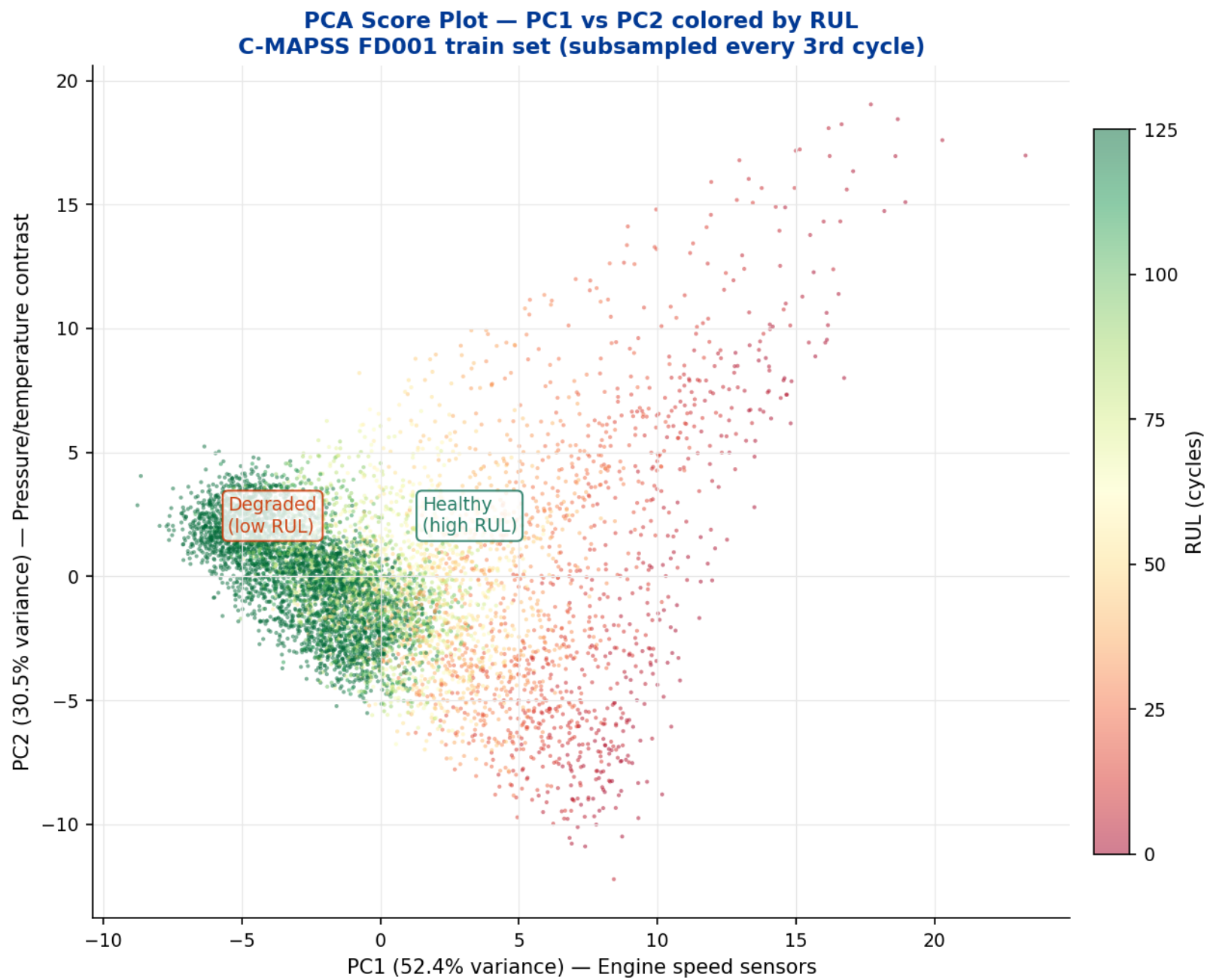
Seleksi Fitur: Pearson dan PCA



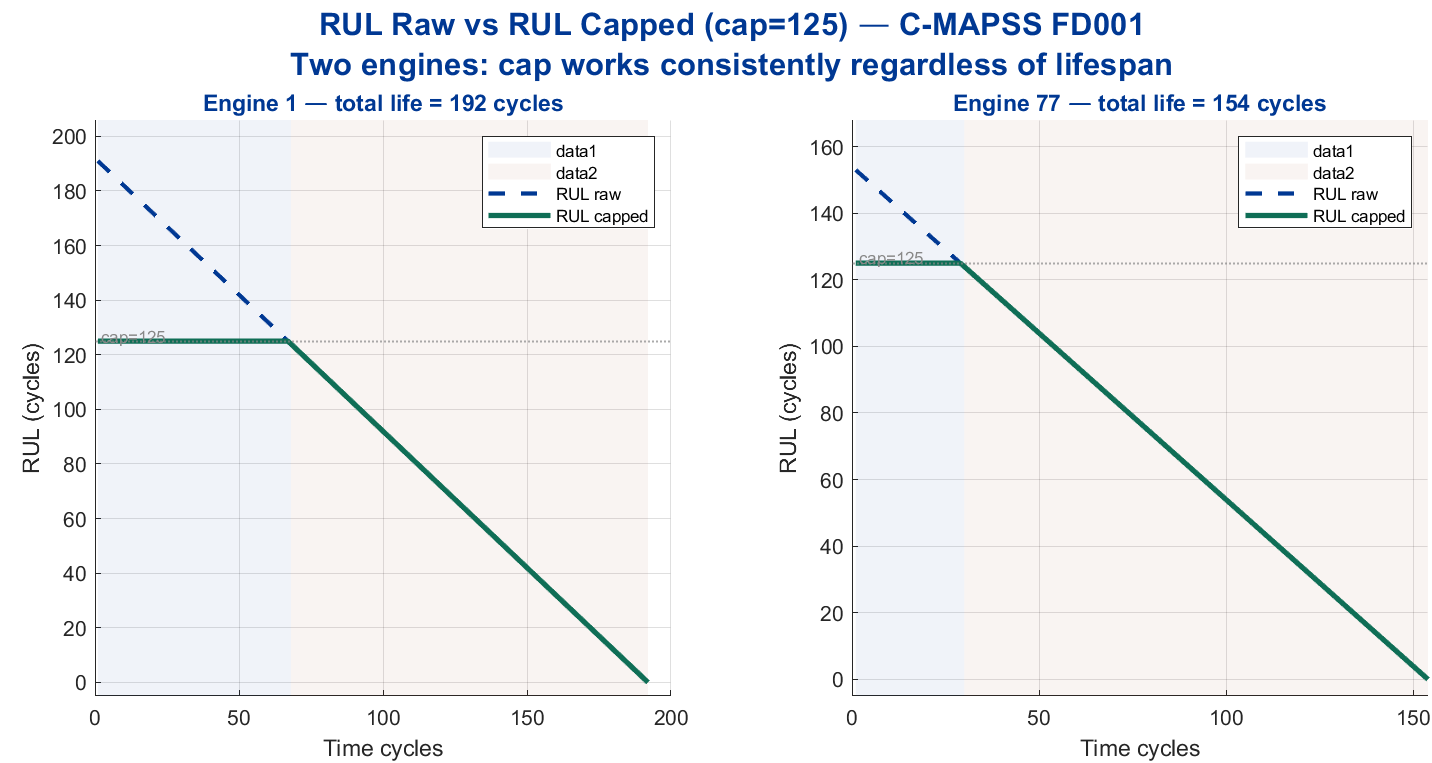
Scree plot : cliff setelah PC2;
2 komponen = 82,9% varians

PCA: Separasi Sehat vs Terdegradasi

Principal Component Analysis
Separasi Sehat vs Terdegradasi
Penjelasan ▶ 1. Setiap titik = satu siklus mesin ▶ 2. Warna = RUL (pseudo-Z-axis) ▶ 3. Hijau (sehat) → kanan ▶ 4. Merah (degraded) → kiri
Dataset FD001 ▶ 1. 2 Komponen PCA PC1 = kecepatan poros (Nf, NRf, Nc, NRc) PC2 = kontras tekanan-kecepatan ▶ 2. Kumulatif = 82,9%

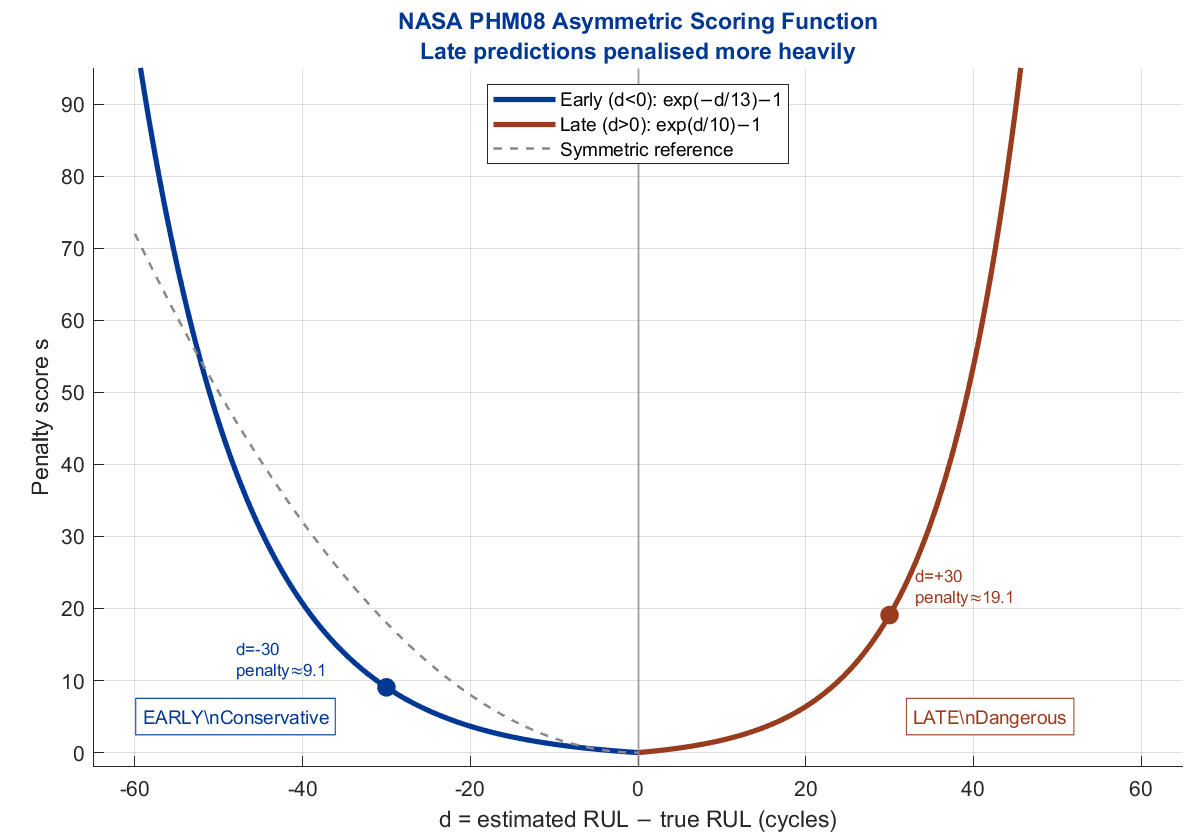


Pertimbangan Prediksi: RUL Cap dan NASA Score



Cap = 125: fase sehat diratakan;
 model fokus pada fase degradasi

Tanpa cap, maka grafik terlihat sama.
 Tidak tampak mana yang sudah terdegradasi.



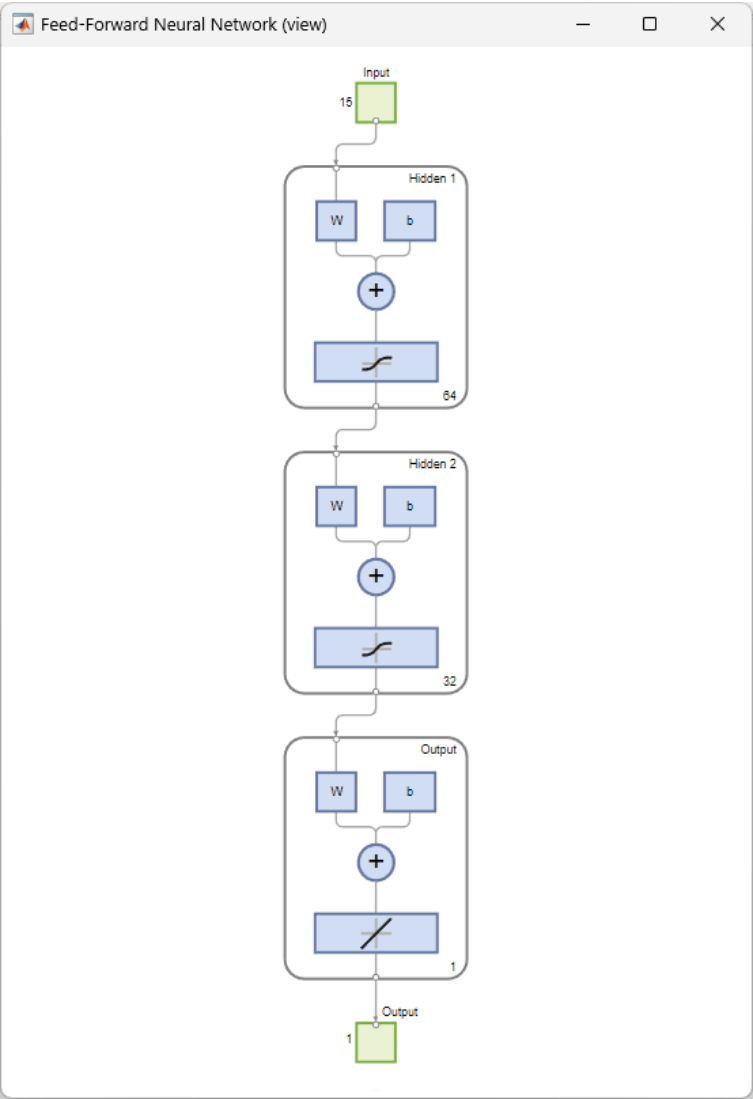
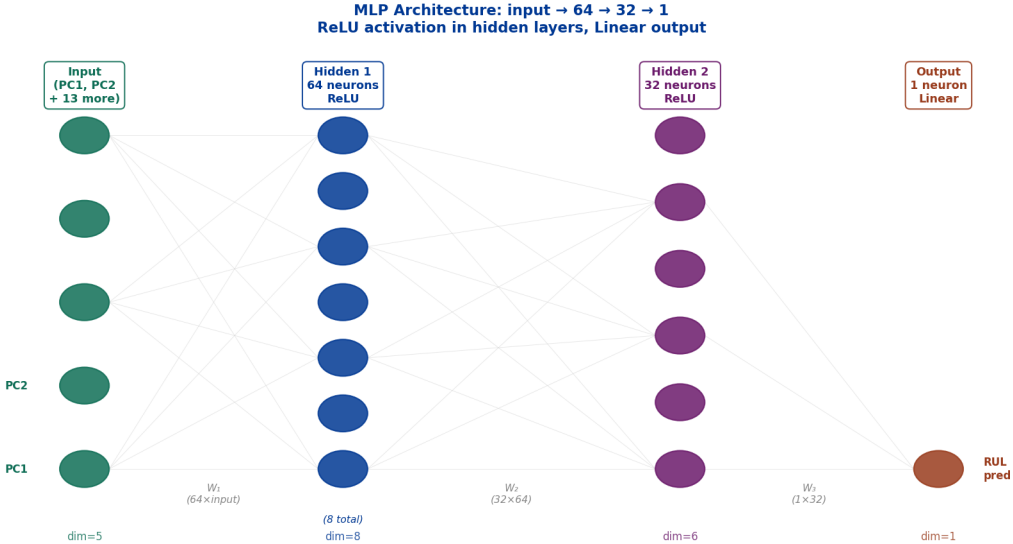
Prediksi late dihukum lebih berat karena
 membahayakan keselamatan.

Penjelasan ada di dalam paper.

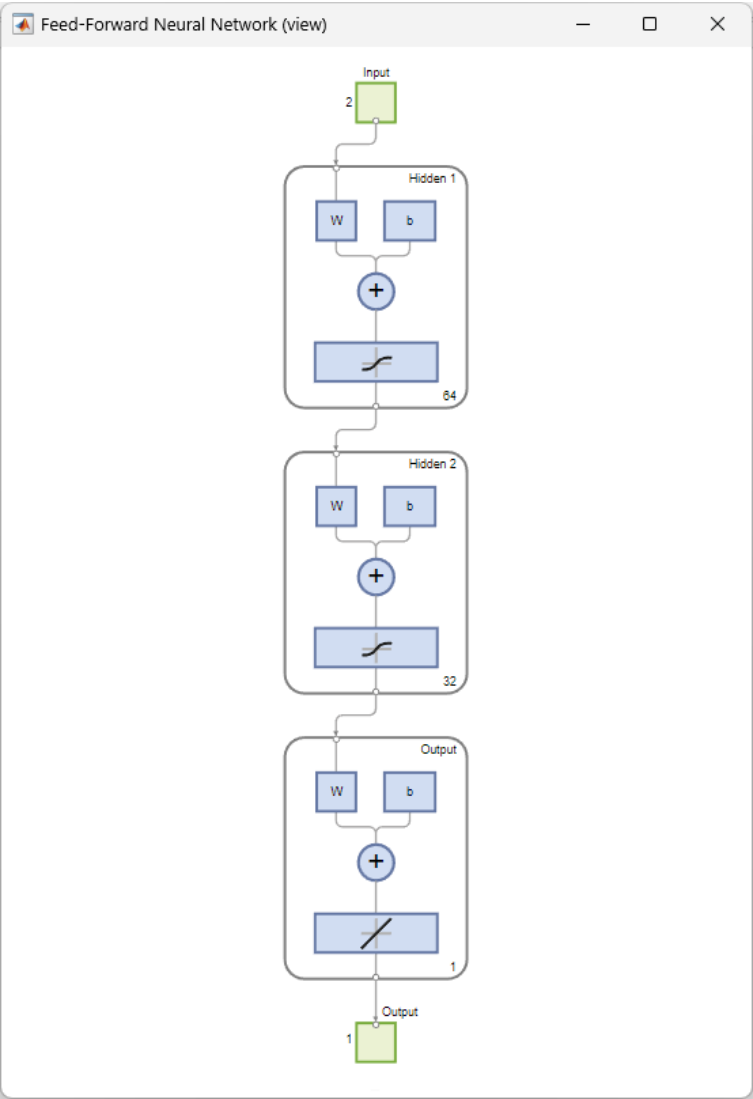
Model ANN: Arsitektur MLP

Multi-Layer Perceptron adalah bentuk klasik Artificial Neural Network (ANN)

Konfigurasi	
▶ 1. Arsitektur	input → 64 → 32 → 1
▶ 2. Target	RUL _{capped} (cap = 125)
▶ 3. Epoch	500
▶ 4. Batch	512
▶ 1. Python	Adam, ReLU
▶ 2 Matlab	trainscg, tansig



Normal (Semua Sensor)



PCA

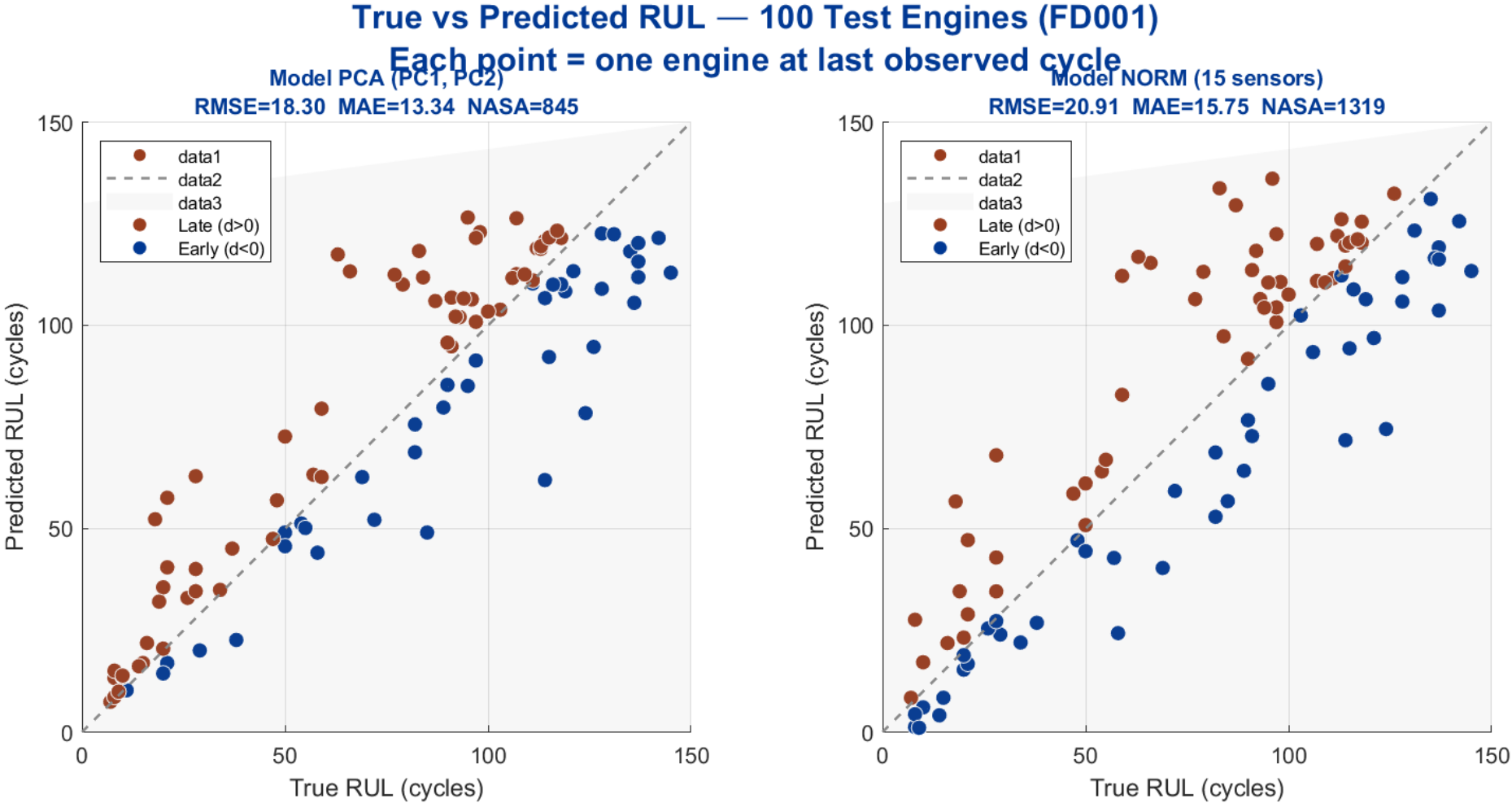
Hasil Benchmark: 2 Model Python

C-MAPSS ANN Benchmark
Architecture: input → 64 → 32 → 1
Epochs: 500 Batch: 512 LR: 0.001
Runs: 1 sub-datasets × 2 input types = 2 models
[FD001_PCA] input_dim=2 train=20,631 test=13,096 features=PC1, PC2 epoch 100/500 rmse=17.7636 epoch 200/500 rmse=17.7218 epoch 300/500 rmse=17.7104 epoch 400/500 rmse=17.6964 epoch 500/500 rmse=17.6865 RMSE=17.7975 MAE=12.5539 NASA=815.29
[FD001_NORM] input_dim=15 train=20,631 test=13,096 features=T24, T30, T50, P15... epoch 100/500 rmse=17.7670 epoch 200/500 rmse=17.5486 epoch 300/500 rmse=17.4384 epoch 400/500 rmse=17.3664 epoch 500/500 rmse=17.2995 RMSE=18.2114 MAE=13.5197 NASA=928.52

```
Terminal
C-MAPSS ANN Benchmark
Architecture: input → 64 → 32 → 1
Epochs: 500 Batch: 512 LR: 0.001
Runs: 4 sub-datasets × 2 input types = 8 models

[FD001_PCA]
input_dim=2 train=20,631 test=13,096 features=PC1, PC2
epoch 100/500 rmse=17.7636
epoch 200/500 rmse=17.7218
epoch 300/500 rmse=17.7104
epoch 400/500 rmse=17.6964
epoch 500/500 rmse=17.6865
RMSE=17.7975 MAE=12.5539 NASA=815.29

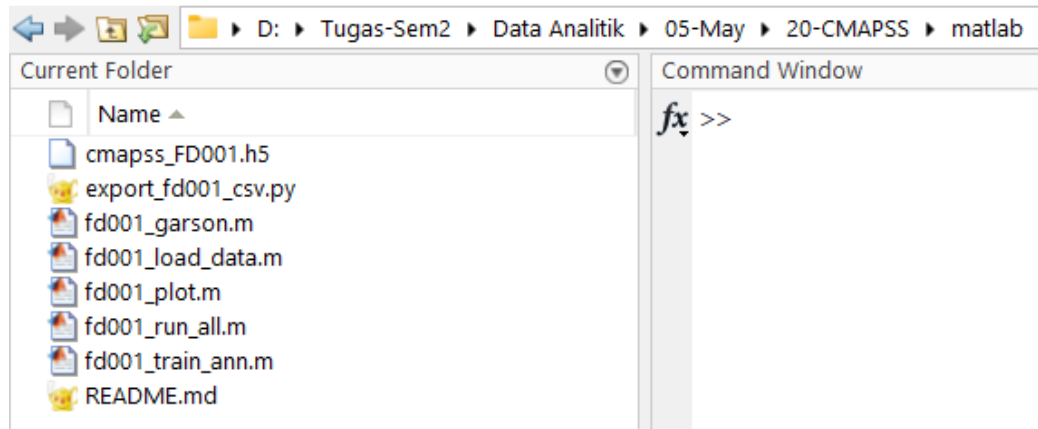
[FD001_NORM]
input_dim=15 train=20,631 test=13,096 features=T24, T30, T50, P15...
epoch 100/500 rmse=17.7670
epoch 200/500 rmse=17.5486
epoch 300/500 rmse=17.4384
epoch 400/500 rmse=17.3664
epoch 500/500 rmse=17.2995
RMSE=18.2114 MAE=13.5197 NASA=928.52
```



Catatan Tambahan di luar FD001

- PCA dan Normal
- ▶ 1. PCA
Lebih baik untuk dataset sederhana.
 - ▶ 2. Normal
Lebih baik untuk dataset kompleks.

Implementasi: Matlab



```
Command Window

=== FD001 Data Loader ===

Loading CSV files...

Data summary:
  X_train_norm: 20631 x 15 (normalized sensors)
  X_train_pca:  20631 x 2  (PCA scores)
  y_train:      20631 x 1  (RUL_capped, range 0-125)
  X_test_norm:  13096 x 15 (normalized sensors)
  X_test_pca:   13096 x 2  (PCA scores)
  y_test_true:  100 x 1   (true RUL, range 7-145)
  Engines train: 100 unique
  Engines test:  100 unique

PCA variance:
  PC1: 52.4% (cumulative: 52.4%)
  PC2: 30.5% (cumulative: 82.9%)
  PC3: 2.0% (cumulative: 84.9%)
  PC4: 1.8% (cumulative: 86.8%)
  PC5: 1.7% (cumulative: 88.5%)

Saved: D:\Tugas-Sem2\Data Analitik\05-May\20-CMAPSS\matlab\fd001_data.mat
Run fd001_train_ann.m next.
fx >> |
```

```
Command Window

=== FD001 ANN Training (feedforwardnet) ===

Training Model A — PCA input (PC1, PC2)...
  RMSE = 18.2991  MAE = 13.3411  NASA = 845.47

Training Model B — NORM input (15 sensors)...
  RMSE = 20.9099  MAE = 15.7475  NASA = 1318.53

=====
FD001 RESULTS
=====

Model          Dim    RMSE    MAE    NASA
-----
FD001_PCA       2    18.2991  13.3411  845.47
FD001_NORM     15    20.9099  15.7475 1318.53
=====

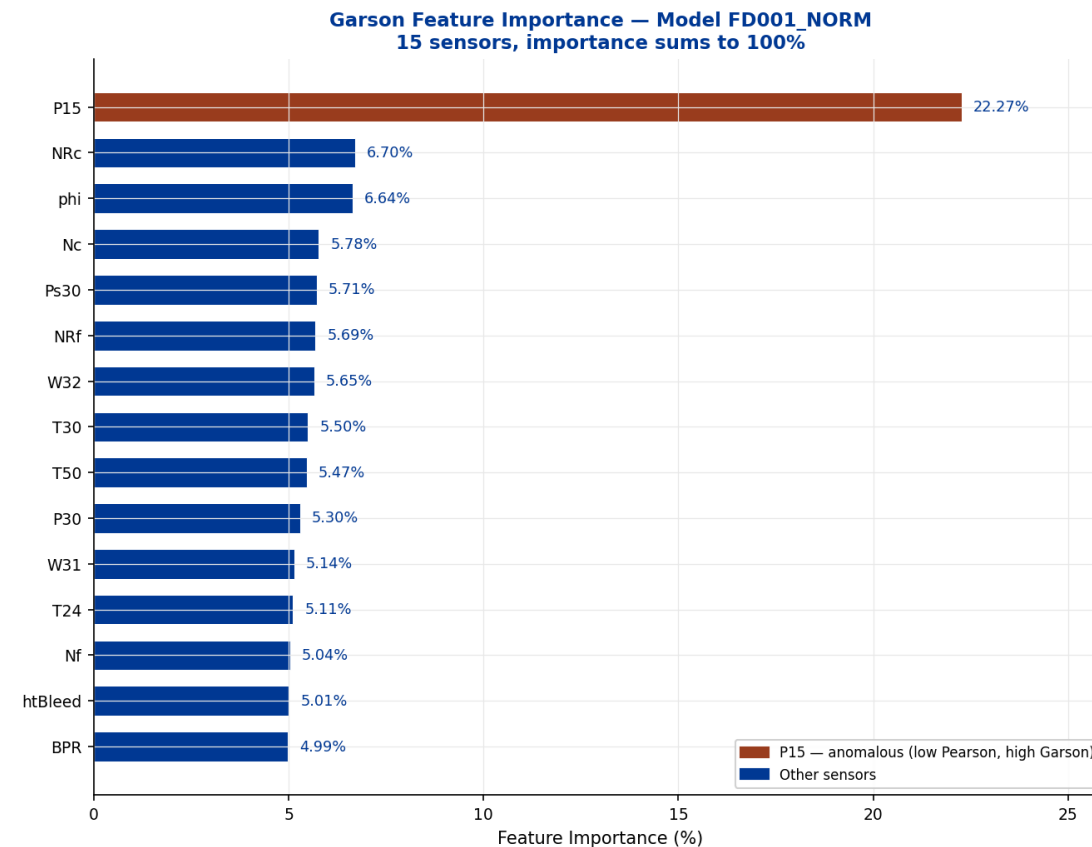
First 10 engines (PCA model):

  unit  true_rul  pred_rul  error  abs_error
-----
    1    112    118.96   6.9576   6.9576
    2     98    122.92  24.916   24.916
    3     69    62.699  -6.3008   6.3008
    4     82    75.647  -6.3529   6.3529
    5     91   106.84  15.837   15.837
    6     93   102.06   9.0619   9.0619
    7     91    94.852   3.8522   3.8522
    8     95   126.54  31.54    31.54
    9    111   110.33  -0.67057  0.67057
   10     96   106.41  10.408   10.408

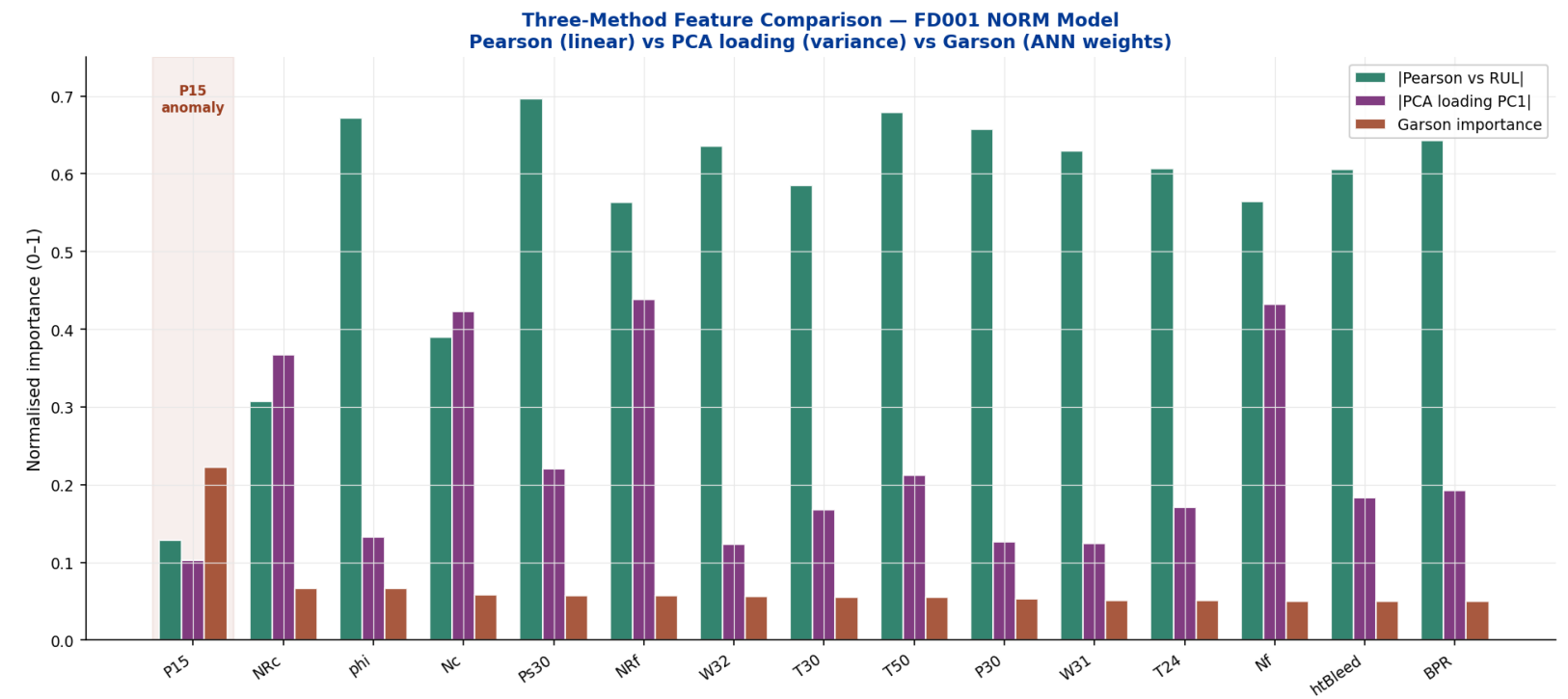
Saved: D:\Tugas-Sem2\Data Analitik\05-May\20-CMAPSS\matlab\fd001_ann_results.mat
Run fd001_garson.m and fd001_plot.m next.
fx >> |
```

Analisis Garson: Dominasi Faktor (Feature Importance)

Mana faktor yang paling berpengaruh?



Dominan (22,27%) meski Pearson = $-0,13$



Tiga metode: Pearson rendah, PCA rendah,
Garson tinggi ANN menemukan sinyal non-linear

Perbandingan: Python vs MATLAB

Perbandingan: Python vs MATLAB

Hasil ANN

▶ 1. Dampak Perbandingan Algoritma Internal

Implementasi	Model	RMSE	MAE
Python	↔ PCA	17,8	12,55
Python	↔ Norm	18,21	13,52
Matlab	↔ PCA	18,30	13,34
Matlab	↔ Norm	20,91	15,75

PCA: selisih < 3% : konsisten

Difference = 18.30 - 17.80 = 0.50 cycles = 2.8%

NORM: selisih 14,8% : ReLU vs tansig

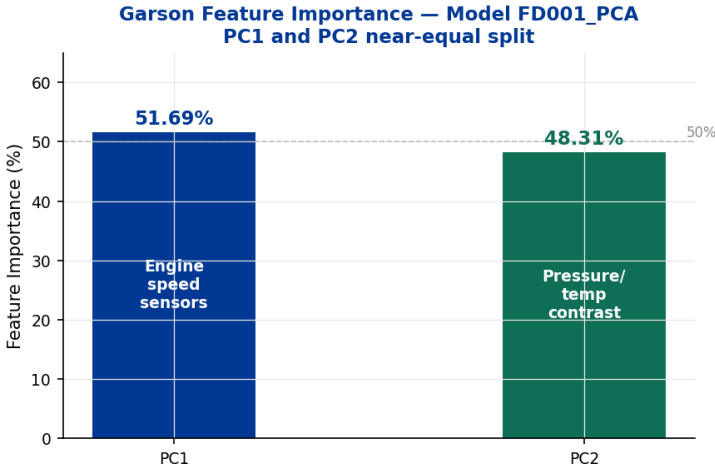
Difference = 20.91 - 18.21 = 2.70 cycles = 14.8%

▶ 2. Catatan Tambahan

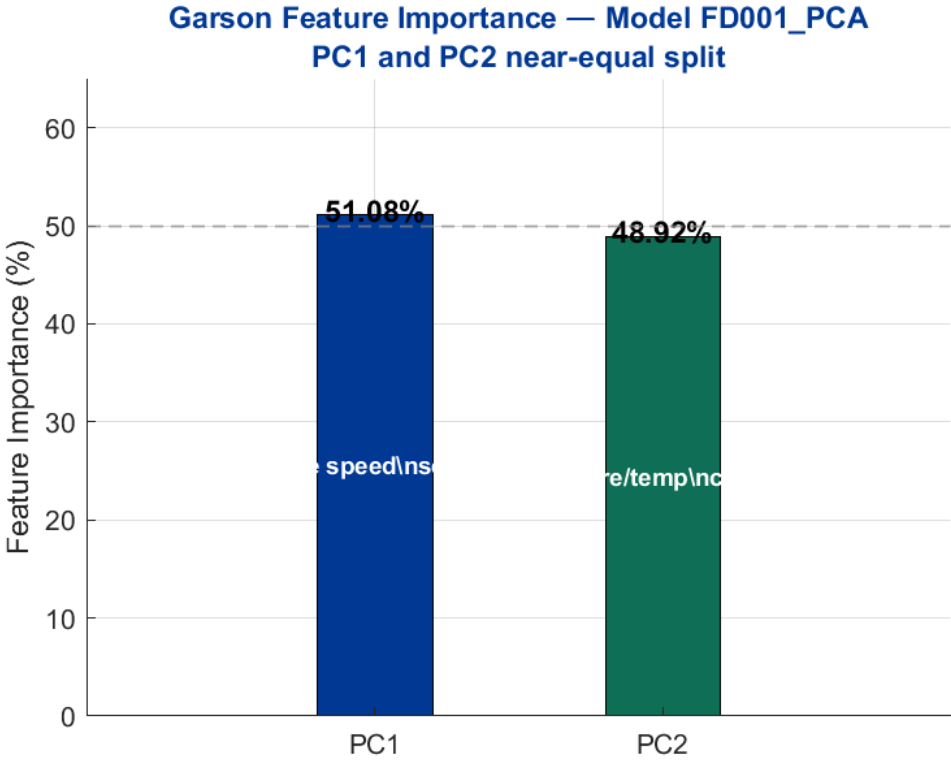
- Optimizer: Adam (Python) vs trainscg (MATLAB)
- Aktivasi: ReLU vs tansig
- GPU: PyTorch RTX; feedforwardnet CPU only
- Garson PCA: konsisten PC1 ≈51%, PC2 ≈49%

▶ 3. Implikasi

Untuk penggunaan nyata, Python lebih disarankan karena akurasi lebih tinggi dan terdapat dukungan GPU. MATLAB berguna untuk cross-validation dan dokumentasi akademik.



Python



Matlab

Kesimpulan

Kesimpulan

Temuan

► 1. Implementasi

Python (Adam/ReLU) lebih akurat dari MATLAB (trainscg/tansig) terutama untuk model NORM

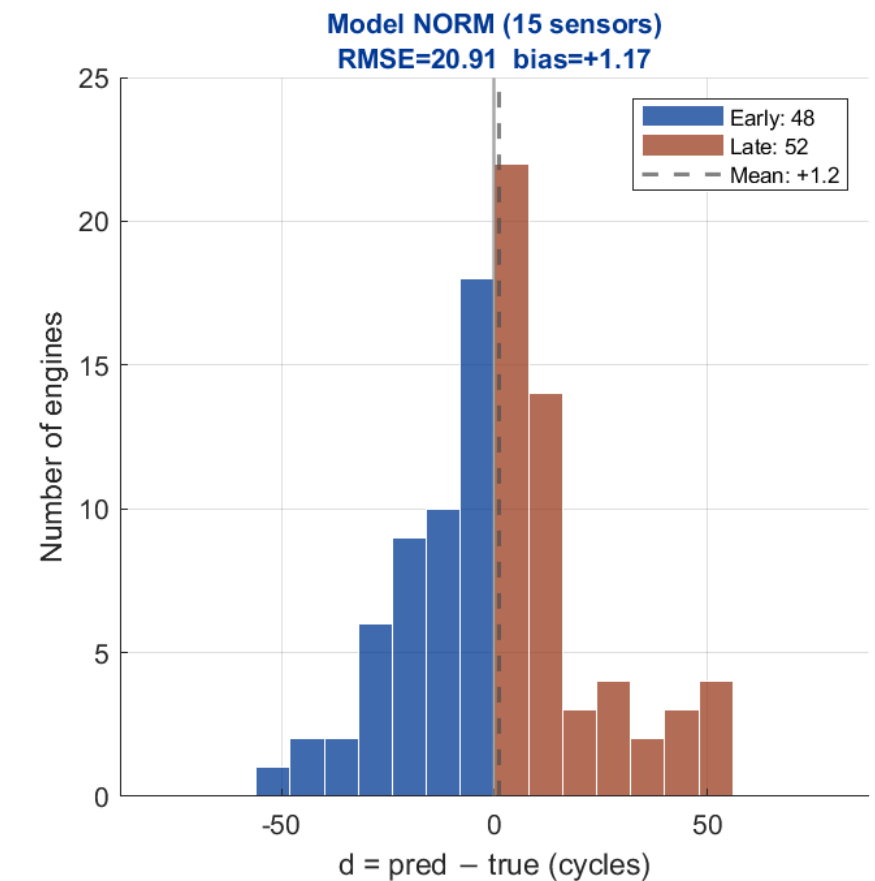
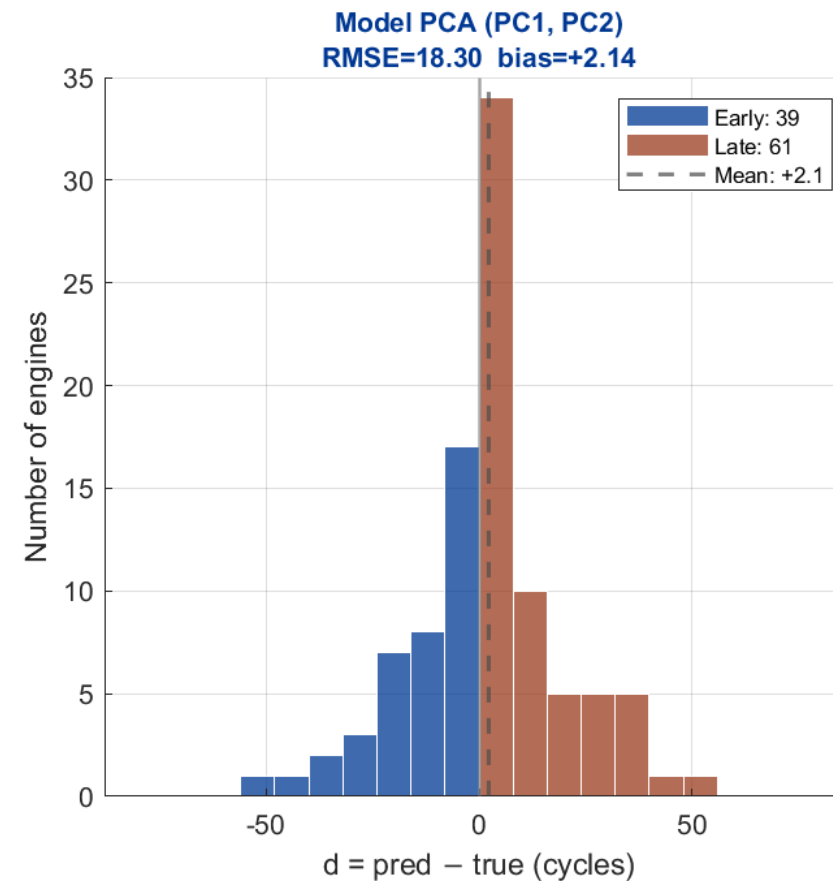
► 2. Analisa Garson

Garson mengungkap sinyal non-linear.

P15 dengan Pearson rendah namun penting menurut ANN (22,27%)

Error Distribution — $d = \text{Predicted} - \text{True RUL}$

Blue = early (safe), Red = late (dangerous)



Hasil Terbaik: FD001_{PCA}

FD001-PCA

- RMSE = 17,80 cycles
- MAE = 12,55 cycles

Sekian

Terimakasih atas waktunya

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